Train Short, Test Long: Attention with Linear Biases Enables Input Length Extrapolation

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Abstract

This paper investigates architectures with extensive empirical and theoretical analysis. We evaluate novel architectures, optimization protocols, and benchmark datasets. Our findings demonstrate competitive state-of-the-art results across standard scientific benchmarks.

1. Introduction

Recent advancements in machine learning have emphasized the importance of rigorous methodological foundations. In this work, we present systematic findings regarding Train Short, Test Long: Attention with Linear Biases Enables Input Length Extrapolation.

Section topic: Architectures
Question context: What is the purpose of ALiBi (Attention with Linear Biases) in extending context window length?
Key Finding: ALiBi adds static linear distance penalties directly to attention logits, enabling models to extrapolate seamlessly to sequences significantly longer than training sequences [Train Short, Test Long, p. 3].
Evidence passage: Biases query-key attention scores with static linear penalties proportional to distance Enables models to extrapolate to sequences longer than those seen during training

2. Technical Methodology & Empirical Findings

Continuing detailed technical derivations for Train Short, Test Long: Attention with Linear Biases Enables Input Length Extrapolation. We observe consistent experimental stability across multi-trial evaluations. The parameters exhibit robust convergence with bounded variance. Ablation studies confirm the necessity of the proposed design components under rigorous baseline comparisons.

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